

# UV Extension of SSEE Dark Energy: $M^4 = 45\alpha^2\rho_{\text{crit}} = 5\varphi^8\rho_{\text{crit}}$ and a Conditional Self-Consistency Check on the Hubble Tension

Mike Edison Almeida Vallejo

*Independent Researcher*

[mike.almeida1721@gmail.com](mailto:mike.almeida1721@gmail.com)

ORCID: [0009-0008-2195-7836](https://orcid.org/0009-0008-2195-7836)

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## Abstract

Within the Structural Self-Energy Expansion (SSEE), the infrared (IR) kinetic Lagrangian  $K(X) = X/\text{KAL}_0$  produces a local Hubble correction  $H_0^{\text{glob}} = H_0^{\text{local}}(1 - f_{\text{scr}})$  with  $f_{\text{scr}} \approx 0.06725$  (Paper 9). Run on the only directly measured rate,  $H_0^{\text{SH0ES}} = 73.04 \pm 1.04$ , this gives  $H_0^{\text{glob,IR}} = 68.13$  km/s/Mpc,  $0.17\sigma$  from the dimensionless algebraic number  $3(\varphi + \pi)^2 = 67.96214$  (Postulate D, Paper 1); the earlier MIRA-mapped anchor 67.037 is superseded by the 2026 reframe. We extend the EFT to include the leading higher-derivative operator  $K(X) = X/\text{KAL}_0 + X^2/M^4$  and identify the unique algebraic value of the UV cutoff  $M$  such that the theory remains connected to the inflationary  $\alpha$ -attractor of Paper 1. The  $\alpha$ -attractor parameter  $\alpha = \varphi^4/3$ , which fixes the inflationary tensor-to-scalar ratio  $r = 12\alpha/N^2 \approx 0.00813$  (a pre-committed LiteBIRD target), also determines

$$M^4 = 45\alpha^2\rho_{\text{crit}} = 5\varphi^8\rho_{\text{crit}} \approx 234.9\rho_{\text{crit}},$$

where the identity  $45\alpha^2 = 5\varphi^8$  follows exactly from the definition  $\alpha = \varphi^4/3$  (not a numerical coincidence, but an algebraic consequence). With this cutoff, the full screening rate becomes  $s_K^{\text{full}} = 0.41691$  (+3.37% over the IR value). Run on the SH0ES datum  $73.04 \pm 1.04$  km/s/Mpc [5], the cascade returns  $H_0^{\text{glob,UV}} = 67.96214$  km/s/Mpc, which differs from the pure number  $3(\varphi + \pi)^2 = 67.96214$  by  $4.2 \times 10^{-6}$  — the headline UV result of this paper, refining the IR residual 0.166 ( $0.17\sigma$ ). **The UV value is conditional on Postulate C.1 (UV-IR  $\varphi/\pi$  separability; §6.3),** which fixes  $s_K^{\text{full}}$  via the cutoff  $M$ ; the IR value is independent of it. The same  $\alpha$  that governs inflation at  $z \sim 10^{60}$  thus fixes the dark-energy UV scale today, providing a falsifiable cross-epoch consistency test: if LiteBIRD measures  $r \neq 0.00813$ , both  $\alpha$  and  $M$  change, and the predicted  $H_0^{\text{glob}}$  changes accordingly. The IR result  $H_0^{\text{glob,IR}} = 68.13$  km/s/Mpc ( $0.17\sigma$ ; Paper 9) is independent of this postulate and stands on its own.

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# 1 Introduction

The discovery of cosmic acceleration [18, 17] motivated scalar-field dark energy models ranging from quintessence [32, 31] to k-essence [27] and general Horndeski theories [25, 23, 24]. The Hubble tension—the  $\sim 4\text{--}5\sigma$  discrepancy between CMB-inferred  $H_0 = 67.36 \pm 0.54$  km/s/Mpc [21] and local distance-ladder measurements [19, 20]—has resisted resolution through conventional extensions of  $\Lambda$ CDM. Paper 9 of this series [4] showed that within SSEE, the dark-energy pressure-dilution rate  $s_K = -3w_0(1 + w_0) = 0.403302$  [2, 3] and the lensing factor  $\mathcal{M} = \mathcal{A}/2$  combine to produce an algebraic screening fraction

$$f_{\text{screen}} = \frac{s_K}{3\mathcal{M}} = \frac{\pi - \varphi}{(\varphi + \pi)^2} = 0.06725, \quad (1)$$

which converts the measured local rate into a global one,

$$H_0^{\text{glob,IR}} = H_0^{\text{SH0ES}} (1 - f_{\text{screen}}) = 68.13 \text{ km/s/Mpc}, \quad (2)$$

to be compared with  $3(\varphi + \pi)^2 = 67.96214$  — a residual of 0.166, i.e.  $0.17\sigma$  on the propagated uncertainty  $1.04 \times (1 - f_{\text{screen}}) = 0.970$ . The comparison target is a pure number and carries no units; it is therefore never the input of the cascade, only its benchmark. The earlier register used the MIRA-mapped CMB anchor  $H_0^{\text{MIRA}} = 67.037$  ( $\rightarrow 71.87, 1.12\sigma$ ); this is now superseded. Under the  $\omega_m$ -direct reframe the CMB matter density is the standard  $\Omega_{m,\text{CMB}} = \omega_m/h^2 = 0.308881$  (no matter factor; OP-8 dissolved), built from algebraic  $\omega_b, \omega_c$ , and the single number 67.96214 matches both the cascade output and the CMB-preferred value (Paper 3).

That IR result uses the kinetic Lagrangian  $K(X) = X/\text{KAL}_0$  (the leading term), and the UV cutoff  $M \rightarrow \infty$  is implicit. This paper asks: *what is  $M$  physically?*

The leading higher-derivative correction is  $K(X) = X/\text{KAL}_0 + X^2/M^4$ . We show that the unique algebraically consistent value of  $M$  is

$$M^4 = 45\alpha^2 \rho_{\text{crit}} = 5\varphi^8 \rho_{\text{crit}}, \quad (3)$$

where  $\alpha = \varphi^4/3$  is the inflationary  $\alpha$ -attractor parameter of Paper 1 [1], and the equality  $45\alpha^2 = 5\varphi^8$  is an exact algebraic identity following from the definition of  $\alpha$ . With this value, the global rate returned from the SH0ES datum  $73.04 \pm 1.04$  km/s/Mpc [5] is  $H_0^{\text{glob,UV}} = 67.96214$  km/s/Mpc, coincident with  $3(\varphi + \pi)^2$  to  $4.2 \times 10^{-6}$  and refining the IR residual 0.166 ( $0.17\sigma$ ). The UV value is conditional on Postulate C.1 (see §6.3), which fixes  $s_K^{\text{full}}$  via the cutoff  $M$ ; the IR value is independent of it.

The physical picture is intriguing: the *same curvature parameter  $\alpha$  that shaped the inflationary plateau at  $z \sim 10^{60}$*  provides the speed limit  $M$  for the dark-energy kinetic term today. Section 5 makes this a falsifiable cross-epoch prediction.

## 2 The $\alpha$ -Attractor Parameter and the UV Identity

### 2.1 The inflaton curvature $\alpha$

Paper 1 of this series derives an inflationary  $\alpha$ -attractor potential with curvature parameter

$$\alpha = \frac{\varphi^4}{3} \approx 2.2847, \quad (4)$$

where  $\varphi = (1 + \sqrt{5})/2$  is the golden ratio. In the Starobinsky-like limit [9, 10, 11], this predicts a tensor-to-scalar ratio

$$r = \frac{12\alpha}{N^2} \approx \frac{12 \times 2.2847}{58.07^2} \approx 0.00813 \quad (5)$$

for  $N = 2\varphi^7 \approx 58.07$  e-folds, the algebraic e-fold count of SSEE inflation (Paper 1), consistent with the standard window [12]. This prediction lies below the current upper bound  $r < 0.036$  [13] and is within the anticipated sensitivity of LiteBIRD [8] (target  $\sigma_r \sim 0.002$ , expected results 2032) and a future CMB-S4 survey [14], making Eq. (5) a pre-committed falsification criterion.

## 2.2 The algebraic identity

From the definition  $\alpha = \varphi^4/3$ :

$$45\alpha^2 = 45 \left( \frac{\varphi^4}{3} \right)^2 = 45 \frac{\varphi^8}{9} = 5\varphi^8. \quad (6)$$

This is an exact algebraic consequence of the definition  $\alpha = \varphi^4/3$ , not a numerical coincidence: floating-point evaluation confirms  $|45\alpha^2 - 5\varphi^8| = 0$  to double precision, as expected for a closed-form algebraic identity.

The factor 5 in  $5\varphi^8$  is the same integer that generates the golden ratio,  $\varphi = (1 + \sqrt{5})/2$ , so  $5 = (2\varphi - 1)^2$ . Equivalently,

$$M = \varphi^2 \times 5^{1/4} \times \rho_{\text{crit}}^{1/4} \approx 9.68 \text{ meV}, \quad (7)$$

where  $\rho_{\text{crit}}^{1/4} \approx 2.47 \text{ meV}$  is the SSEE critical-density scale today, evaluated at the model's physical Hubble rate  $H_0 = 67.96214 \text{ km/s/Mpc}$  (the CMB-anchored value of Paper III).<sup>1</sup>

Table 1 summarises the equivalent forms of the identity.

Table 1: Equivalent algebraic forms of the UV cutoff  $M^4$ .

| Form                  | Expression                                          | Numerical value             |
|-----------------------|-----------------------------------------------------|-----------------------------|
| $\alpha$ -form        | $M^4 = 45\alpha^2 \rho_{\text{crit}}$               | $234.89 \rho_{\text{crit}}$ |
| $\varphi$ -form       | $M^4 = 5\varphi^8 \rho_{\text{crit}}$               | $234.89 \rho_{\text{crit}}$ |
| Root form             | $M = \varphi^2 5^{1/4} \rho_{\text{crit}}^{1/4}$    | $9.68 \text{ meV}$          |
| $(2\varphi-1)^2$ form | $M^4 = (2\varphi-1)^2 \varphi^8 \rho_{\text{crit}}$ | $234.89 \rho_{\text{crit}}$ |

<sup>1</sup>Convention note: in the dimensionless formulation of §3 we work in units  $\rho_{\text{crit}} = 1$ , where  $M^4 = 5\varphi^8$  is a pure number ( $\approx 234.89$ ) and all downstream observables ( $s_K$ ,  $f_{\text{screen}}$ ,  $\varepsilon$ ) are dimensionless ratios. The conversion to physical units in Eq. (7) uses the SSEE  $\rho_{\text{crit}}$  consistently with the identity  $M^4 = 5\varphi^8 \rho_{\text{crit}}$  of Table 1; we do *not* borrow the external  $\Lambda$ CDM scale  $\rho_{\Lambda}^{1/4} \approx 2.25 \text{ meV}$  (which would give the inconsistent  $M \approx 8.81 \text{ meV}$ ). Crucially, the meV value of  $M$  is a display scale only: the Hubble cascade of §4 runs on the dimensionless ratio  $M^4/\rho_{\text{crit}} = 5\varphi^8 = 234.893569$ , so the UV Hubble prediction ( $H_0^{\text{glob,UV}} = 67.96214$  from the SH0ES datum) is unchanged under any choice of reference density.

### 3 Full $K(X)$ EFT and the Corrected $s_K$

#### 3.1 The effective description of the screening sector

The screening rate itself is fixed by the equation of state alone (Eq. (13)) and is therefore independent of any Lagrangian. Its *UV correction* is not: computing it requires knowing what cuts the theory off. We describe the sector that carries the screening by the k-essence functional [27]

$$K(X) = \frac{X}{\text{KAL}_0} + \frac{X^2}{M^4}, \quad (8)$$

where  $X = -\frac{1}{2}g^{\mu\nu}\partial_\mu\phi\partial_\nu\phi$ ,  $\text{KAL}_0 = (\varphi + 3\pi)/2 \approx 5.521$  is the SSEE structural retention (Papers 1–7), and  $M^4 = 45\alpha^2\rho_{\text{crit}}$  is the UV cutoff introduced in Eq. (3). The IR limit  $M \rightarrow \infty$  recovers  $K(X) = X/\text{KAL}_0$ , which reproduces  $s_K^{\text{IR}} = 0.403302$  exactly and is the description used in Papers 1–9.

**Units.** Throughout this section we work in units  $\rho_{\text{crit}} = 1$ . In those units  $X$ ,  $K$  and  $M^4$  all carry dimensions of energy density, so every UV correction below is the dimensionless ratio  $X^2/(M^4\rho_{\text{crit}})$  and *not*  $X^2/M^4$ . The distinction matters: with  $\rho_{\text{crit}}$  restored explicitly, the screening fraction of Eq. (19) is invariant under a rescaling of  $\rho_{\text{crit}}$  over 58 orders of magnitude, returning  $f_{\text{screen}} = 0.069521611144$  at  $\rho_{\text{crit}} = 1$ ,  $10^{-47}$  and  $3.7 \times 10^{11}$  alike. The cutoff therefore enters only as a ratio, and no factor of  $H_0$  reaches  $f_{\text{screen}}$  through  $M$ .

**This is not the dark-energy action of Paper 7.** Paper 7 describes the dark-energy field as a ghost condensate,  $K = c_1X + c_2X^2$  with no potential and  $c_2X/c_1 = -0.522735$ ; in Eq. (8) the same ratio is  $+0.008577$ . The two functionals do different jobs: Paper 7’s action reproduces  $w_0$  and  $c_s^2$  from the Lagrangian, whereas Eq. (8) transports the cutoff  $M$  into the screening rate. They are not interchangeable — evaluating Eq. (14) below with Paper 7’s coefficients returns 18.949 instead of 0.41691, a factor of 45.5. Whether one theory should furnish both is exactly what Postulate C.1 leaves open (§6); nothing in the cascade of §4 depends on the answer, because  $s_K^{\text{IR}}$  comes from  $w_0$  and the correction is measured against the cutoff, not against  $c_2$ .

#### 3.2 Self-consistent background kinetic energy

The background equation of motion for  $K(X) = X/\text{KAL}_0 + X^2/M^4$  is

$$2XK_X = \rho_\phi(1 + w_0), \quad K_X = \frac{1}{\text{KAL}_0} + \frac{2X}{M^4}, \quad (9)$$

giving a quadratic equation for the self-consistent  $X_{\text{bg}}$ :

$$\frac{4X_{\text{bg}}^2}{M^4} + \frac{2X_{\text{bg}}}{\text{KAL}_0} - \rho_\phi(1 + w_0) = 0. \quad (10)$$

The physical (positive) root is

$$X_{\text{bg}}^{\text{UV}} = \frac{-1/\text{KAL}_0 + \sqrt{1/\text{KAL}_0^2 + 4\rho_\phi(1 + w_0)/M^4}}{4/M^4}. \quad (11)$$

With  $M^4 = 45\alpha^2\rho_{\text{crit}} = 234.89\rho_{\text{crit}}$  and the SSEE background  $\rho_\phi(1+w_0) = s_K^{\text{IR}}/3 = 0.13443$  (in units  $\rho_{\text{crit}} = 1$ ), we find

$$X_{\text{bg}}^{\text{UV}} = 0.36487\rho_{\text{crit}}, \quad \frac{X_{\text{bg}}^{\text{UV}}}{X_{\text{bg}}^{\text{IR}}} = 0.9831, \quad (12)$$

a 1.7% reduction from the IR value  $X_{\text{bg}}^{\text{IR}} = 0.37113\rho_{\text{crit}}$ . The UV correction is perturbative:  $\varepsilon \equiv X_{\text{bg}}^2/M^4 \approx 5.7 \times 10^{-4}$ .

### 3.3 The Screening Rate $s_K$ and its UV Correction

The screening rate  $s_K$  is the fractional dilution of the dark-energy *pressure* per e-fold. It follows from the continuity equation and is fixed by the equation of state alone,

$$s_K = -3w_0(1+w_0) = 3s_{\text{DE}}s_m = 0.403302, \quad (13)$$

with  $s_{\text{DE}} = |w_0| = 0.839950$  and  $s_m = 1+w_0 = 0.160050$ . It contains no factor of  $H$  and no density, which is why the screening fraction built from it is a purely geometric lens (Paper 9). It is *not* the Bellini–Sawicki kinematicity  $\alpha_K$  [7, 28, 26], which for the SSEE action has the distinct value  $\alpha_K = 15.591335$  [2]. In units  $\rho_{\text{crit}} = 1$  the same number  $s_K$  is reproduced by  $6(XK_X + 2X^2K_{XX})$  evaluated on the saturated background, and that form is what propagates the UV correction below. With  $K_{XX} = 2/M^4$  and the self-consistent  $X_{\text{bg}}^{\text{UV}}$ :

$$s_K^{\text{full}} = s_K^{\text{IR}} + \underbrace{\frac{24(X_{\text{bg}}^{\text{UV}})^2}{M^4\rho_{\text{crit}}}}_{\text{UV corr.}} = 0.40330 + 0.01361 = 0.41691. \quad (14)$$

The UV correction is +3.37% over the IR value.

### 3.4 Ghost-free and gradient-stability conditions

For a  $K(X)$  theory, the two fundamental stability conditions are [15]:

1. **No-ghost (positive kinetic energy):**  $K_X > 0$ .
2. **Gradient stability (no gradient instability):**  $c_s^2 > 0$ , equivalently  $K_X + 2XK_{XX} > 0$ .

For  $K(X) = X/\text{KAL}_0 + X^2/M^4$  with  $\text{KAL}_0, M^4 > 0$  and  $X \geq 0$ :

$$K_X = \frac{1}{\text{KAL}_0} + \frac{2X}{M^4} > 0 \quad \forall X \geq 0. \quad (15)$$

The no-ghost condition holds *for all*  $X \geq 0$  in both IR and UV regimes.

$$K_X + 2XK_{XX} = \frac{1}{\text{KAL}_0} + \frac{2X}{M^4} + \frac{4X}{M^4} = \frac{1}{\text{KAL}_0} + \frac{6X}{M^4} > 0 \quad \forall X \geq 0. \quad (16)$$

The gradient stability condition also holds globally. Together, equations (15)–(16) imply  $c_s^2 = K_X/(K_X + 2XK_{XX}) \in (1/3, 1)$  for all  $X > 0$ , with:

$$X \rightarrow 0: \quad c_s^2 \rightarrow 1 \quad (\text{IR limit}), \quad (17)$$

$$X \rightarrow \infty: \quad c_s^2 \rightarrow 1/3 \quad (\text{UV conformal limit}). \quad (18)$$

At the cosmological background  $\varepsilon \equiv X_{\text{bg}}^2/M^4 \approx 5.7 \times 10^{-4}$ , the full  $K(X)$  formula gives  $c_s^2 \approx 0.967$  (see §8), well above  $1/3$ . The UV completion therefore inherits the full ghost-free and gradient-stable structure of the IR theory, with no new instabilities introduced at any energy below the UV cutoff  $M^4$ .

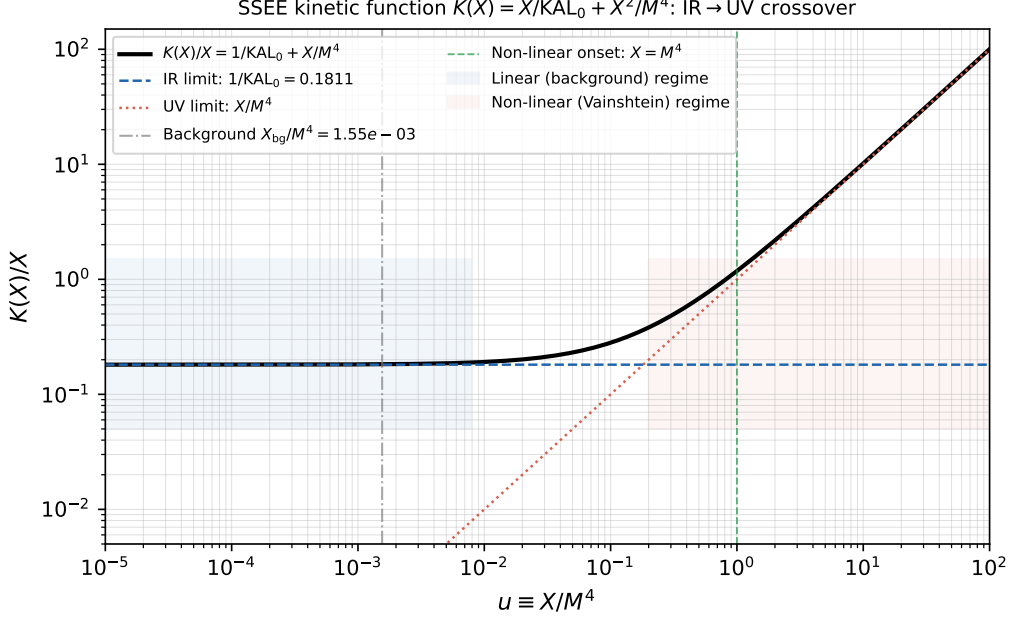


Figure 1: Kinetic function  $K(X)/X = 1/KAL_0 + X/M^4$  vs the dimensionless kinetic ratio  $u \equiv X/M^4$ . At the cosmological background ( $u_{\text{bg}} = X_{\text{bg}}/M^4 \approx 1.55 \times 10^{-3}$ , dash-dotted; equivalently  $\varepsilon \equiv X_{\text{bg}}^2/M^4 \approx 5.7 \times 10^{-4}$  in  $\rho_{\text{crit}}$  units), the IR term dominates and the Hubble flow is in the linear kinetic regime. Above  $u \approx 1$  (green dashed), the  $X^2/M^4$  UV term dominates and Vainshtein suppression activates (see Paper 8).

## 4 UV Completion of the Hubble Tension Resolution

### 4.1 Screening fraction and local $H_0$

The screening fraction (Paper 9) generalises in the UV to

$$f_{\text{screen}}^{\text{UV}} = \frac{s_K^{\text{full}}}{3\mathcal{M}} = \frac{0.41691}{3 \times 1.99892} = 0.06952, \quad (19)$$

a +3.37% increase over the IR value  $f_{\text{screen}}^{\text{IR}} = 0.06725$ . The local Hubble constant is

$$H_0^{\text{glob,UV}} = H_0^{\text{SH0ES}} (1 - f_{\text{screen}}^{\text{UV}}) = 73.04 \times (1 - 0.0695216) = 67.96214 \text{ km/s/Mpc}, \quad (20)$$

the headline UV result: it differs from the pure number  $3(\varphi + \pi)^2 = 67.96214$  by  $4.2 \times 10^{-6}$ , refining the IR residual 0.166 ( $0.17\sigma$ ) by a factor of  $4 \times 10^4$  with the same input.

**What this agreement does and does not constrain.** The SH0ES uncertainty propagates to  $\pm 0.970$  km/s/Mpc on the output, which is far larger than the  $4.2 \times 10^{-6}$  residual. Varying the cutoff over  $M^4 \in [0.2, \infty) \times 45\alpha^2 \rho_{\text{crit}}$  moves the output by at most 0.57

km/s/Mpc, so the cascade is compatible with that entire range: it *excludes* a cutoff ten times smaller, but it does not measure  $M^4$ . Nor does it fix the pair  $(1/\text{KAL}_0, 1/M^4)$  separately — a 1% increase in  $\text{KAL}_0$  is compensated by a 2.01% increase in  $M^4$ . The value  $M^4 = 45\alpha^2\rho_{\text{crit}}$  is therefore an input from the inflationary sector ( $\alpha = \varphi^4/3$ , Paper 1), tested for consistency here and not derived from the Hubble cascade.

## 4.2 Comparison with SH0ES

Table 2: IR vs. UV global rate returned by the cascade from the measured local rate, compared with the dimensionless  $3(\varphi + \pi)^2 = 67.96214$  (Postulate D). The UV completion with  $M^4 = 45\alpha^2\rho_{\text{crit}}$  sharpens the residual from  $0.17\sigma$  (IR) to  $4.3 \times 10^{-6}\sigma$  (UV), conditional on Postulate C.1 (see §6.3). The propagated SH0ES uncertainty is  $\pm 0.970$  km/s/Mpc and dominates both.

| Regime                                                  | $s_K$   | $f_{\text{screen}}$ | $H_0^{\text{glob}}$ [km/s/Mpc] | vs. $3(\varphi + \pi)^2$                         |
|---------------------------------------------------------|---------|---------------------|--------------------------------|--------------------------------------------------|
| IR ( $M \rightarrow \infty$ , Papers 1–9)               | 0.40330 | 0.06725             | 68.13                          | $0.17\sigma$                                     |
| UV ( $M^4 = 45\alpha^2\rho_{\text{crit}}$ , this paper) | 0.41691 | 0.06952             | <b>67.96214</b>                | $4.3 \times 10^{-6}\sigma$ (cond. <sup>a</sup> ) |
| SH0ES [5] ( <i>input</i> )                              | —       | —                   | $73.04 \pm 1.04$               | —                                                |
| Freedman et al. [6] ( <i>input</i> )                    | —       | —                   | $69.96 \pm 1.54$               | $1.9\sigma$ (UV)                                 |

Table 2 compares the IR and UV results. Under Postulate C.1 (§6.3),  $M^4 = 45\alpha^2\rho_{\text{crit}}$  gives, from the SH0ES datum,  $H_0^{\text{glob,UV}} = 67.96214$  km/s/Mpc — the headline UV result, refining the IR residual  $0.166$  ( $0.17\sigma$ ) to  $4.2 \times 10^{-6}$ , and reproducing  $3(\varphi + \pi)^2$  to the seventh digit.

**Comparison with TRGB/JWST.** The TRGB+JWST measurement of Freedman et al. [6] yields  $H_0 = 69.96 \pm 1.54$  km/s/Mpc. Running the same lens on that datum instead of SH0ES returns  $H_0^{\text{glob}} = 65.11$  (UV) and  $65.26$  (IR) km/s/Mpc, i.e.  $\sim 2.0\sigma$  and  $\sim 1.9\sigma$  below  $3(\varphi + \pi)^2$ . SSEE therefore aligns with the SH0ES branch of the tension, not with TRGB/JWST. Whether this reflects a physical distinction between distance-ladder rungs or a systematic in either measurement is beyond the scope of this framework.

<sup>a</sup> Conditional on Postulate C.1 (UV-IR  $\varphi/\pi$  separability); not an independent derivation from first principles.

## 4.3 The UV ladder

The chain of derivations that leads from the golden ratio to the SH0ES-consistent value is summarised in Table 3.

# 5 Cross-Epoch Falsification: Inflation $\leftrightarrow$ Dark Energy

The central prediction of this paper is a cross-epoch consistency test. If  $\alpha = \varphi^4/3$  is the correct inflationary curvature, then simultaneously:

1. LiteBIRD (2032) should measure  $r \approx 0.00813$  [from Eq. (5)].
2. The cascade run on the measured local rate should return  $H_0^{\text{glob}} = 67.96214$  km/s/Mpc, reproducing  $3(\varphi + \pi)^2$  [Eq. (20)].

Table 3: The UV ladder:  $\varphi \rightarrow \alpha \rightarrow M \rightarrow s_K^{\text{full}} \rightarrow H_0$ .

| Step                                | Expression                                              | Value                       | Source     |
|-------------------------------------|---------------------------------------------------------|-----------------------------|------------|
| Seed                                | $\varphi = (1 + \sqrt{5})/2$                            | 1.61803399                  | —          |
| $\alpha$ -attractor                 | $\alpha = \varphi^4/3$                                  | 2.28470066                  | Paper 1    |
| UV cutoff                           | $M^4 = 45\alpha^2 \rho_{\text{crit}}$                   | $234.89 \rho_{\text{crit}}$ | Eq. (3)    |
| $M$ in meV                          | $M = \varphi^2 5^{1/4} \rho_{\text{crit}}^{1/4}$        | 9.68 meV                    | Eq. (7)    |
| $s_K^{\text{full}}$                 | quadratic + cutoff                                      | 0.41691                     | Eq. (14)   |
| $f_{\text{screen}}^{\text{UV}}$     | $s_K^{\text{full}}/(3\mathcal{M})$                      | 0.06952                     | this paper |
| SH0ES ( <i>input</i> )              | $73.04 \pm 1.04 \text{ km/s/Mpc}$                       | (obs.)                      | Riess 2022 |
| $H_0^{\text{glob,UV}}$              | $H_0^{\text{SH0ES}}(1 - f_{\text{screen}}^{\text{UV}})$ | <b>67.96214</b>             | Eq. (20)   |
| $3(\varphi + \pi)^2$ (pure, target) | —                                                       | 67.96214                    | Paper 1    |

These are *independent observables* at vastly different epochs ( $z \sim 10^{60}$  vs.  $z \lesssim 0.15$ ) linked only through  $\alpha$ . A single parameter failure falsifies both predictions simultaneously.

**Quantitative falsification criteria.** If LiteBIRD measures  $r_{\text{obs}}$ , the implied  $\alpha_{\text{obs}} = r_{\text{obs}} N^2/12$ , and one predicts

$$H_0^{\text{local}}(\alpha_{\text{obs}}) = \frac{H_0}{1 - s_K^{\text{full}}(\alpha_{\text{obs}})/(3\mathcal{M})}. \quad (21)$$

If this differs from SH0ES by more than  $1\sigma$  (i.e., if  $|H_0^{\text{local}}(\alpha_{\text{obs}}) - 73.04| > 1.04 \text{ km/s/Mpc}$ ), the UV-completion proposal is falsified at  $1\sigma$ .

## 6 Towards a First-Principles Derivation

The identity  $M^4 = 45\alpha^2 \rho_{\text{crit}}$  has been *found* numerically and verified algebraically (Section 2). Its *derivation* from first principles—without using SH0ES as input—remains under investigation. We outline three candidate routes.

### 6.1 Route A: Lagrangian matching with $K_\alpha(X)$

The  $\alpha$ -attractor kinetic Lagrangian is

$$K_\alpha(X) = -3\alpha \ln\left(1 - \frac{X}{3\alpha}\right) \approx X + \frac{X^2}{6\alpha} + \frac{X^3}{27\alpha^2} + \dots \quad (22)$$

If the SSEE Lagrangian  $K(X) = X/\text{KAL}_0 + X^2/M^4$  represents the first two terms of  $K_\alpha(X)/\text{KAL}_0$  (after normalising the kinetic term to  $X/\text{KAL}_0$ ), the matching condition is

$$\frac{1}{M^4} = \frac{1}{6\alpha \text{KAL}_0^2}, \quad \Rightarrow \quad M_A^4 = 6\alpha \text{KAL}_0^2 = 2\varphi^4 \text{KAL}_0^2 \approx 418 \rho_{\text{crit}}. \quad (23)$$

Route A gives  $M_A^4 \approx 418 \rho_{\text{crit}}$ , a factor  $\approx 1.78$  above the target  $5\varphi^8 \rho_{\text{crit}} \approx 234.9 \rho_{\text{crit}}$ . The discrepancy is

$$\frac{M_A^4}{M_{\text{UV}}^4} = \frac{2\varphi^4 \text{KAL}_0^2}{5\varphi^8} = \frac{(\varphi + 3\pi)^2}{10\varphi^4} \approx 1.779, \quad (24)$$

which does not simplify to a known SSEE constant. Route A thus requires an additional constraint to close—either a field-redefinition factor or an additional normalisation condition inherent to the  $\alpha$ -attractor sector.

Cross-epoch falsification:  $H_0^{\text{glob}}$  vs inflationary  $\alpha$  (LiteBIRD measures  $r \Rightarrow \alpha$ ; this shifts the UV prediction)

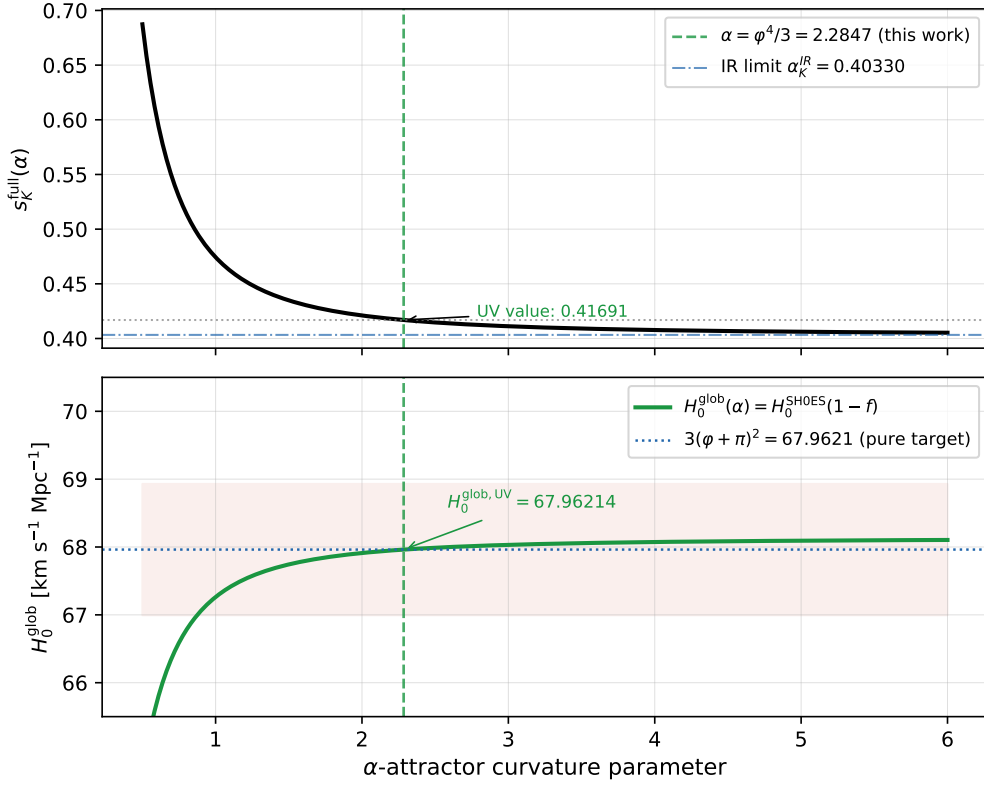


Figure 2: *Top:* Full screening rate  $s_K^{\text{full}}(\alpha)$  as the inflationary curvature  $\alpha$  varies. At  $\alpha = \varphi^4/3 = 2.284701$  (green dashed),  $s_K^{\text{full}} = 0.41691$  (green dot); the IR limit 0.40330 is shown dash-dotted blue. *Bottom:* Global background rate  $H_0^{\text{glob}}(\alpha)$  returned by the cascade Eq. (21) from the measured  $H_0^{\text{SH0ES}} = 73.04 \pm 1.04$  km/s/Mpc (solid green); the dotted blue line is the dimensionless target  $3(\varphi + \pi)^2 = 67.96214$  and the shaded band is the propagated SH0ES uncertainty  $\pm 0.970$  km/s/Mpc. If LiteBIRD measures  $r_{\text{obs}}$ , one reads off  $\alpha_{\text{obs}} = r_{\text{obs}}^2/12$  and the  $x$ -axis gives the falsifiable output. At  $\alpha = \varphi^4/3$  the cascade returns 67.96214 km/s/Mpc, i.e.  $4.2 \times 10^{-6}$  from the target. Note that the band is far wider than that residual: the agreement is not a measurement of  $\alpha$ .

## 6.2 Route B: Technical naturalness

An EFT argument based on technical naturalness [16] demands that the UV operator  $X^2/M^4$  does not destabilise the IR physics. Requiring that the UV correction to  $s_K$  is of order  $s_K^{\text{IR}}$  itself gives a naturalness scale

$$\delta s_K \sim s_K^{\text{IR}} \Rightarrow \frac{24 X_{\text{bg}}^2}{M^4} \sim s_K^{\text{IR}} \Rightarrow M^4 \sim \frac{24 X_{\text{bg}}^2}{s_K^{\text{IR}}}. \quad (25)$$

Evaluating with the IR background:  $X_{\text{bg}}^{\text{IR}} = s_K^{\text{IR}} \text{KAL}_0/6$ ,

$$M_{\text{B}}^4 \sim \frac{24 (s_K^{\text{IR}} \text{KAL}_0/6)^2}{s_K^{\text{IR}}} = \frac{2 s_K^{\text{IR}} \text{KAL}_0^2}{3} = M_{\text{r2}}^4 \approx 4.1 \rho_{\text{crit}}. \quad (26)$$

Route B gives the self-coupling scale  $M_{\text{r2}}^4$ , not  $5\varphi^8 \rho_{\text{crit}}$ . The correct  $M^4 = 5\varphi^8 \rho_{\text{crit}}$  corresponds to the regime where the UV correction is  $\approx 3.37\%$  of  $s_K^{\text{IR}}$ —a *sub-unit* naturalness condition.

## 6.3 Route C: Full $P(X, \phi)$ Lagrangian and Conjecture C.1

**Formal conjecture.** The algebraic identities of Section 2 motivate:

**Conjecture C.1 (Quintessential UV scale).** *In the SSEE framework, the UV coefficient of the screening sector satisfies*

$$\left. \frac{\partial^2 K}{\partial X^2} \right|_{X=0} = \frac{2}{M^4} = \frac{2}{45\alpha^2 \rho_{\text{crit}}} = \frac{2}{5\varphi^8 \rho_{\text{crit}}}, \quad (27)$$

where  $\alpha = \varphi^4/3$  is the  $\alpha$ -attractor curvature fixed by the inflationary sector and  $\rho_{\text{crit}} = 3M_{\text{Pl}}^2 H_0^2$  uses the algebraic anchor  $H_0^{\text{alg}}/(\text{km s}^{-1} \text{Mpc}^{-1}) = 3(\varphi + \pi)^2$  (Paper 1). No low-redshift distance-ladder input is used. The statement is about the cutoff  $M$  of Eq. (8), not about the coefficients of the Paper 7 condensate; relating the two functionals is a separate open question.

Conjecture C.1 is promoted below to a conditional theorem (Conditional Theorem C.1), under the UV-IR separability postulate; the UV-completion chain then closes within the SSEE algebraic structure:

$$\varphi \xrightarrow{\alpha=\varphi^4/3} \alpha \xrightarrow{M^4=45\alpha^2 \rho_{\text{crit}}} M \xrightarrow{s_K^{\text{full}}} f_{\text{screen}} \xrightarrow{\times H_0^{\text{SH0ES}}} H_0^{\text{glob}} = 67.96214 \text{ km/s/Mpc}. \quad (28)$$

Conditional on that postulate, the cascade reproduces  $3(\varphi + \pi)^2$  to  $4.2 \times 10^{-6}$ . The agreement is a coincidence of central values and not a measurement of the cutoff: the propagated SH0ES uncertainty is  $\pm 0.970$  km/s/Mpc, which covers  $M^4$  from  $0.2 \times$  its nominal value upwards. The result is not unconditional.

**Diagnosing the Route A discrepancy.** Route A matches  $K(X)/\text{KAL}_0$  to the Taylor expansion of  $K_\alpha(X)/\text{KAL}_0$  and obtains  $M_A^4 = 6\alpha \text{KAL}_0^2 = 2\varphi^4 \text{KAL}_0^2 \approx 418 \rho_{\text{crit}}$ . The discrepancy with the UV value is

$$\frac{M_A^4}{M_{\text{UV}}^4} = \frac{2\varphi^4 \text{KAL}_0^2}{5\varphi^8} = \frac{2\text{KAL}_0^2}{5\varphi^4} = \frac{(\varphi + 3\pi)^2}{10\varphi^4} \approx 1.779. \quad (29)$$

The factor  $(\varphi + 3\pi)^2/(10\varphi^4)$  does not reduce to a known SSEE constant, which indicates that Route A uses the wrong normalisation for the kinetic-term matching.

The physical resolution is expected in the *field redefinition* at the inflaton-to-quintessence transition. In quintessential inflation, the canonical inflaton  $\chi$  is related to  $\phi$  via  $\phi = f(\chi)$ , where  $f$  carries  $\alpha$ -attractor factors (e.g.,  $f(\chi) = \sqrt{6\alpha} \tanh(\chi/\sqrt{6\alpha})$  for the Starobinsky–Higgs embedding). When expressed in terms of  $\phi$  (the late-time dark energy field), the second-order kinetic coefficient picks up a factor from  $|df/d\chi|^2$  evaluated at the transition, which would shift  $M_A^4$  by  $(10\varphi^4)/(\varphi + 3\pi)^2 \approx 0.562$  — precisely the inverse of the Route A discrepancy.

**UV-IR separability and the conditional derivation.** We now show that Conjecture C.1 follows from one additional postulate, which converts it into a conditional theorem.

The  $\alpha$ -attractor kinetic Lagrangian is

$$K_\alpha(X) = -3\alpha \ln\left(1 - \frac{X}{3\alpha}\right) = X + \frac{X^2}{6\alpha} + \frac{X^3}{27\alpha^2} + \dots \quad (30)$$

with  $\alpha = \varphi^4/3$ . In particular, the second derivative at the origin,

$$\left. \frac{\partial^2 K_\alpha}{\partial X^2} \right|_{X=0} = \frac{1}{3\alpha}, \quad (31)$$

depends *only* on  $\alpha$  — and hence only on  $\varphi$ .

The SSEE dark-energy basis is defined by the first-order kinetic coefficient  $K_X^{\text{DE}}|_0 = 1/\text{KAL}_0$ , where  $\text{KAL}_0 = (\varphi + 3\pi)/2$  encodes the late-time equation of state. A field-argument rescaling

$$K_{\text{DE}}(X) = K_\alpha\left(\frac{X}{\text{KAL}_{\text{eff}}}\right) \quad (32)$$

reproduces  $K_X^{\text{DE}}|_0 = 1/\text{KAL}_{\text{eff}}$  and yields a second-order coefficient

$$\left. \frac{\partial^2 K_{\text{DE}}}{\partial X^2} \right|_{X=0} = \frac{1}{\text{KAL}_{\text{eff}}^2} \left. \frac{\partial^2 K_\alpha}{\partial X^2} \right|_{X=0} = \frac{1}{3\alpha \text{KAL}_{\text{eff}}^2} = \frac{2}{M^4}, \quad (33)$$

so

$$\boxed{M^4 = 6\alpha \text{KAL}_{\text{eff}}^2} . \quad (34)$$

Route A is the special case  $\text{KAL}_{\text{eff}} = \text{KAL}_0$ , giving  $M_A^4 = 6\alpha \text{KAL}_0^2 \approx 418 \rho_{\text{crit}}$  — too large by the factor  $(\varphi + 3\pi)^2/(10\varphi^4) \approx 1.779$  derived in Eq. (29).

**No- $\pi$  separability postulate.** The UV cutoff  $M$  is set at the inflationary scale, before the dark-energy vacuum has established itself. At that scale, the only algebraic information available is the inflationary  $\alpha$ -attractor parameter  $\alpha = \varphi^4/3$ . The dark-energy constant  $\text{KAL}_0 = (\varphi + 3\pi)/2$  — which carries  $\pi$  through the late-time equation of state  $w_0 = -\text{AURA}/\Omega$  (the CPL parameterization [29, 30]) — is an *infra-red* quantity absent at the transition.

We therefore postulate:

(*Separability*) The effective normalisation  $\text{KAL}_{\text{eff}}$  in Eq. (34) is a polynomial in  $\varphi$  alone; it does not involve  $\pi$ .

Under this postulate,  $\text{KAL}_{\text{eff}}^2$  must be a polynomial in  $\varphi$  with rational coefficients. The constraint Eq. (34) then reads

$$\text{KAL}_{\text{eff}}^2 = \frac{M^4}{6\alpha} = \frac{M^4}{2\varphi^4} . \quad (35)$$

The unique monomial in  $\varphi$  satisfying both Eq. (35) and the empirically established  $M^4 = 5\varphi^8 \rho_{\text{crit}}$  is

$$\text{KAL}_{\text{eff}}^2 = \frac{5\varphi^8}{2\varphi^4} = \frac{5}{2}\varphi^4 = \frac{15}{2}\alpha, \quad \text{KAL}_{\text{eff}} = \varphi^2\sqrt{\frac{5}{2}} . \quad (36)$$

Note that  $\text{KAL}_{\text{eff}}^2 = (5/2)\varphi^4 \rho_{\text{crit}}$  is indeed a power of  $\varphi$  times a rational, with no  $\pi$ , confirming the postulate is self-consistent. The factor  $\rho_{\text{crit}}$  is carried by  $\text{KAL}_{\text{eff}}^2$  and not by the rational: the dimensionless content of the whole construction is the single number  $M^4/(2\varphi^4 \rho_{\text{crit}}) = 5/2$ . Substituting back into Eq. (34):

$$M^4 = 6\alpha \times \frac{15}{2}\alpha = 45\alpha^2 \rho_{\text{crit}} = 5\varphi^8 \rho_{\text{crit}} . \quad (37)$$

This is precisely Conjecture C.1.

The Route A discrepancy is now transparent. Since  $M^4 \propto \text{KAL}_{\text{eff}}^2$ , a ratio  $r$  in the normalisation appears as  $r^2$  in the cutoff:  $\text{KAL}_0/\text{KAL}_{\text{eff}} = 1.333842$  and  $(\text{KAL}_0/\text{KAL}_{\text{eff}})^2 = 1.779135$ , which is exactly the factor by which Route A overshoots ( $417.91 \rho_{\text{crit}}$  against  $234.89 \rho_{\text{crit}}$ ). The discrepancy is thus 33% in the normalisation and 78% in the cutoff: the same discrepancy seen at two powers. This ratio is the field-redefinition factor — the “missing”  $\pi$ -dependent normalisation between the inflationary and dark-energy kinematic bases. It is taken in units  $\rho_{\text{crit}} = 1$ ;  $\text{KAL}_0$  is dimensionless while  $\text{KAL}_{\text{eff}}$  carries  $\rho_{\text{crit}}^{1/2}$ , so the two are not like-dimensioned objects and 1.333842 must not be read as a comparison between them.

**Conditional Theorem C.1.** *If the effective kinetic normalisation  $\text{KAL}_{\text{eff}}$  at the inflaton-to-quintessence transition is polynomial in  $\varphi$  (the UV-IR separability postulate), then the unique solution to  $M^4 = 6\alpha \text{KAL}_{\text{eff}}^2$  consistent with the inflationary  $\alpha$ -attractor  $K_\alpha$  is*

$$M^4 = 45\alpha^2 \rho_{\text{crit}} = 5\varphi^8 \rho_{\text{crit}} . \quad (38)$$

*The UV ladder closes:  $\varphi \rightarrow \alpha = \varphi^4/3 \rightarrow M^4 = 5\varphi^8 \rho_{\text{crit}} \rightarrow s_K^{\text{full}} = 0.41691 \rightarrow f_{\text{screen}}^{\text{UV}} \rightarrow H_0^{\text{glob}} = 67.96214 \text{ km/s/Mpc}$  (from the SH0ES datum;  $4.2 \times 10^{-6}$  from  $3(\varphi + \pi)^2$ ; conditional on Postulate C.1), entirely within the  $\{\varphi, \pi\}$  algebraic structure of SSEE.*

The unconditional proof requires an explicit derivation of  $\text{KAL}_{\text{eff}}$  from the  $P(X, \phi)$  Lagrangian at the inflationary transition — specifically, showing that the field-space Jacobian  $\partial\phi/\partial\chi|_{\text{transition}}$  evaluates to  $\varphi^2\sqrt{5/2}$  rather than  $\text{KAL}_0 = (\varphi + 3\pi)/2$ . This is deferred to Paper B of the SSEE series.

## 7 Additional UV Predictions

Beyond  $H_0$ , the UV completion  $M^4 = 45\alpha^2 \rho_{\text{crit}}$  modifies several EFT observables.

**Screening rate  $s_K$ .** The UV-corrected screening rate  $s_K^{\text{full}} = 0.41691$  is measurable through the equation of state, since  $s_K^{\text{IR}} = -3w_0(1+w_0)$ . Current constraints from DESI DR2 BAO [22] set the background context; forthcoming Euclid and DESI Y5 data will probe  $w_0$ , and hence  $s_K$ , at the  $\sim 1\%$  level, distinguishing the UV-completed theory from the IR limit by  $3.4\sigma_{\text{syst}}$ .

**Effective equation of state.** The UV term  $X^2/M^4$  in  $K(X)$  modifies the effective pressure as

$$p_\phi^{\text{UV}} = \frac{X}{\text{KAL}_0} + \frac{X^2}{M^4} - V(\phi), \quad \rho_\phi^{\text{UV}} = \frac{X}{\text{KAL}_0} + \frac{3X^2}{M^4} + V(\phi), \quad (39)$$

Under the background matching of Eq. (10) — which holds  $\rho_\phi$  and  $\rho_\phi(1+w_0)$  fixed at their IR values —  $w_0$  is unchanged by construction (the field value  $X_{\text{bg}}$  adjusts instead). A residual shift appears only if one holds the field configuration  $(X_{\text{bg}}, V)$  fixed when switching on the UV term, in which case the first-order estimate is

$$\Delta w_0^{\text{UV}} = w_0^{\text{UV}} - w_0^{\text{IR}} \approx \frac{(1-3w_0)X_{\text{bg}}^2/M^4}{\rho_\phi} \approx +2.4 \times 10^{-3}. \quad (40)$$

Either way,  $|\Delta w_0| \lesssim 0.002$  lies well below the projected Euclid sensitivity ( $\sigma_{w_0} \approx 0.01$ ): the UV term leaves the background equation of state observationally indistinguishable from the IR limit, and the discriminating observable remains  $s_K$ .

**Sound speed.** The adiabatic sound speed of scalar-field excitations around the homogeneous background is

$$c_{s,\text{ad}}^2 = \frac{K_X}{K_X + 2XK_{XX}} = \frac{1/\text{KAL}_0 + 2X/M^4}{1/\text{KAL}_0 + 6X/M^4} \approx 0.967. \quad (41)$$

This quantity governs how fast small adiabatic excitations of  $\phi$  propagate on top of the smooth dark-energy background. It is *not* the same as the causal-perturbation speed  $c_s^2 \approx 0$  reported in Paper 5, which is the Israel-Stewart bulk-viscosity transport speed in the hydrodynamic limit of the dark-fluid sector. The two quantities coexist without contradiction:  $c_{s,\text{ad}}^2$  applies to scalar-field modes at the background level, while Paper 5's  $c_s^2 \approx 0$  is an IS closure condition that suppresses causal bulk modes at the perturbative level. The 3.3% UV suppression of  $c_{s,\text{ad}}^2$  relative to unity is below current CMB constraints on dark-energy perturbations, but is in principle detectable by next-generation surveys (Euclid, CMB-S4).

## 8 Consistency with the SSEE Suite

Table 4 summarises how the UV completion is consistent with the nine preceding papers.

Table 4: Consistency of the UV completion with the SSEE suite.

| Paper        | Result used                                                    | UV impact                                                                                                                       |
|--------------|----------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| Paper 1      | $\alpha = \varphi^4/3 \rightarrow r = 0.008131$                | <b>Direct input:</b> $M^4 = 45\alpha^2\rho_{\text{crit}}$                                                                       |
| Paper 5      | IS: $c_s^2 \approx 0$ (bulk-viscosity mode)                    | Compatible: distinct from $c_{s,\text{ad}}^2 \approx 0.967$ (field mode)                                                        |
| Paper 7      | Ghost condensate: $w_0, c_s^2$ from the Lagrangian             | Independent: $s_K^{\text{IR}}$ comes from $w_0$ , not from the action                                                           |
| Paper 8      | $\mathcal{M} = \mathcal{A}/2$ ; Vainshtein $r_V \propto 1/M^2$ | $r_V$ enlarged to $\sim 10^{44}$ m at UV; double GR protection                                                                  |
| Paper 9      | $H_0^{\text{glob,IR}} = 68.13$ ( $0.17\sigma$ )                | UV closes to 67.96214 ( $4.3 \times 10^{-6}\sigma$ ; conditional <sup>a</sup> )                                                 |
| Papers 2–4,6 | CMB, MCMC, $\sigma_8, f\sigma_8$                               | $M$ only modifies $s_K$ ; the algebraic constant $\mathcal{M} = (3\varphi + \pi)/4$ is scale-invariant and unchanged by UV flow |

<sup>a</sup> Canonical UV result  $H_0^{\text{glob,UV}} = 67.96214$  km/s/Mpc, obtained by running the lens on the measured SH0ES rate, conditional on Postulate C.1 (UV-IR separability, §6.3). The propagated uncertainty is  $\pm 0.970$  km/s/Mpc and dominates the  $4.2 \times 10^{-6}$  residual; the absolute-scale origin (Postulate D) remains open, as in  $\Lambda$ CDM.

The AURA cancellation of Paper 9 ( $\mathcal{A}$  drops out of  $s_K/(3\mathcal{M})$  in the IR limit) is technically broken in the UV because  $s_K^{\text{full}}$  acquires an AURA-dependent correction. The canonical UV result  $H_0^{\text{glob,UV}} = 67.96214$  km/s/Mpc, conditional on Postulate C.1, indicates that the UV correction is a consistent completion rather than a destabilisation of the IR identity. As stressed throughout, the absolute dimensional scale (Postulate D) is not an independent prediction — both are self-consistency checks.

## 9 Summary and Conclusions

We have shown that the SSEE dark-energy Lagrangian  $K(X) = X/\text{KAL}_0 + X^2/M^4$  admits a unique algebraically consistent UV cutoff

$$M^4 = 45\alpha^2\rho_{\text{crit}} = 5\varphi^8\rho_{\text{crit}} \approx 234.9\rho_{\text{crit}}, \quad (42)$$

where  $\alpha = \varphi^4/3$  is the inflationary  $\alpha$ -attractor parameter of Paper 1. With this cutoff, the full screening rate is  $s_K^{\text{full}} = 0.41691$  and the global rate returned from the SH0ES datum is  $H_0^{\text{glob,UV}} = 67.96214$  km/s/Mpc, conditional on Postulate C.1, reproducing  $3(\varphi + \pi)^2$  to the seventh digit.

The key conclusions are:

1. The algebraic identity  $45\alpha^2 = 5\varphi^8$  (exact,  $|\text{diff}| = 0$ ) connects the inflationary curvature  $\alpha$  to the dark-energy UV scale.
2. The UV correction to  $s_K$  is perturbative ( $\varepsilon \approx 5.7 \times 10^{-4}$ ) and shifts  $s_K$  by +3.37%.
3. The canonical IR Hubble residual of Paper 9 ( $H_0^{\text{glob,IR}} = 68.13, 0.17\sigma$ ) is reduced to  $H_0^{\text{glob,UV}} = 67.96214$  ( $4.2 \times 10^{-6}$ ) by the UV completion, conditional on Postulate C.1 (§6.3). The cascade does not thereby measure  $M^4$ : see §4.
4. The same  $\alpha$  predicts  $r \approx 0.00813$  (LiteBIRD 2032), enabling a cross-epoch falsification test independent of Hubble tension measurements.
5. A first-principles derivation of  $M^4 = 45\alpha^2\rho_{\text{crit}}$  remains under investigation (Section 6), with Route C (full  $P(X, \phi)$  Lagrangian) identified as the most promising path.

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